ASR-Aware Unified Fusion of Three Speech Enhancers for Noise Event Removal on Vaani

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Team Hydra, IndoML Datathon 2026, Track 2: Noise Event Removal

***Abstract*—Track 2 removes noise events from Indic field speech, scored by SI-SDR plus 100× the word error rate reduction (ΔWER) of the organisers' SraVaani-1.0 ASR. Cleaner waveforms alone do not give the ASR back its words, so our main novelty (boxed in Section II) is to train speech enhancers *through the frozen scoring ASR's own decoder* (encoder distance and transcript likelihood, also on 12–18 s recordings matching the test clips' length) and to fuse three public enhancers of different families (MossFormerGAN, a 104M band-split RNN, TF-Locoformer) per time-frequency cell with a deletion-weighted word-error loss. Every clip takes the same path. The final entry scores Combined 19.897 (SI-SDR 15.793 dB, ΔWER $+4.104$ pp) on the Phase 2 board.**

I. Training Data and Preprocessing

The task and the organiser data. A test clip is either *synthetic* (clean speech with noise events pasted in, scored by scale-invariant signal-to-distortion ratio, SI-SDR, and ΔWER) or *natural* (a field recording, scored by ΔWER). The validation release has 695 synthetic pairs (clean and noisy, 4.5 s median) and 1,806 natural recordings with transcripts, in 27 languages. A fixed hash of the speaker id sets 30% of speakers aside: their 209 pairs never train the models we evaluate with (the *held-out protocol*, 486 training pairs), giving honest scores and unseen inputs for the fusion; our final MossFormerGAN and BSRNN are retrained on all 695 (TF-Locoformer stays on 486: retraining it on all pairs scored lower on the board). Natural clips are never trained on; a fixed 30% hash sample of them (555 clips, 7,834 words) measures natural ΔWER.

Noise events. Noisy minus clean at the annotated spans gives 2,213 events (23 min, 7 categories), used only to build training mixtures.

External speech. Test clips are long (13 s median) single-speaker recordings, while the validation pairs are 4.5 s from few speakers, so we needed more clean speech: Vaani-transcription-part [1], the corpus the competition clips come from, and IndicVoices-R [2]. From them we cut clean *beds* (recordings onto which noise events are pasted) that pass Silero-VAD [3] and TorchAudio-SQUIM [4] checks (Table I).

Data use and fairness. The competition allows external data when disclosed; Table I lists all of it. (a) *No competition recordings in external data.* Vaani-transcription-part is the competition's source corpus, so before any use we deleted every clip named like a competition clip (validation, Track 1 training, Phase 1 test: 98,653 names) and every clip from the same recording as a validation or Phase 1 test clip (9,931 recordings). Phase 2 test clips are anonymised, so we compared content instead: the ASR transcripts of all 3,178 test clips against those of all external speech we trained on (1,450 long beds and all 5,000 fusion-set clips). No test clip shares 30% of its 8-word sequences with any of them (the largest, 20%, is a speaker counting). This rules out reused recordings, though a speaker could appear in both. (b) *No test audio in training*; the only test information used is distribution-level (clip lengths and blind SNR estimates), to size our simulated clips. (c) *Noise events.* Synthetic clips share a common event pool, so test clips may contain events from the validation release we train on; a control model trained without the held-out pairs' events scored the same on them (18.64 vs 18.64 dB after 4k steps, with identical curves throughout), so the models do not rely on having heard a specific event. Nothing at inference stores, matches or subtracts events.

TABLE I
Every dataset used and its role (beds: clean recordings onto which events are pasted). No test audio, external noise or TTS speech is used.

Data (source)	Used	How	Final
Organiser data (Vaani validation release)			
Synthetic pairs	695 pairs, 1.0 h	member training (209 held out)	yes
Noise events (noisy − clean)	2,213, 0.4 h	training mixtures only	yes
Natural recordings	555 of 1,806	evaluation only	no
External data (public, gated); every competition clip and recording removed first			
Vaani-transcription-part [1]	1,450 long beds, 5.7 h	long stream, all members	yes
	710 beds, 1.4 h	fusion mixtures (908 clips)	yes
IndicVoices-R [2]	2,793 beds, 6.5 h	fusion mixtures (1,592 clips)	yes
	198 h more	member training (Table II)	no
Simulated from the above (noise events pasted onto clean speech)			
Fusion set	2,500 clips, 9.9 h	fusion training (153 dev)	yes
Screening set	1,200 clips, 4.2 h	comparing choices; fusion training	yes
Long set	360 clips, 1.3 h	evaluation only	no

Making training examples. A training crop (2 s for MossFormerGAN, 4 s otherwise) is a real pair with probability 0.5, else a clean reference plus 1–3 random events at an SNR drawn from the real pairs. Each step also adds a *long stream* micro-batch of 12–18 s long beds with events pasted as the organisers do (Poisson 0.5 events/s, log-normal gain, median 0.93, σ=0.8), for the ASR losses only. For all training audio the ASR losses' target is SraVaani's own greedy transcript of the clean speech. Three simulated sets: the *fusion set* joins 1–4 beds of one speaker into 8–18 s clips on beds no member saw, to train the fusion; the *screening set* glues the 209 held-out pairs' clean references (reused across clips) into 1,200 test-like clips (33.8k words), to compare training choices; the *long set* pastes events onto one long Vaani recording from each of 360 speakers kept out of all training beds, for evaluation.

Novelty. Prior work trains enhancers to match a recogniser's features (mimic loss [5]), adds part of the noisy input back to limit ASR-harmful artefacts [6], [7], and trains recognisers, not enhancers, with sampled minimum-word-error (MWER) losses [8], [9]. This work contributes:

1) *Enhancers trained on the scoring ASR's own decoder.* The frozen SraVaani's encoder distance and its transducer likelihood of the transcript (with a forward recursion we wrote for words cut at crop edges) improve ΔWER by 0.24–0.61 pp on all three model families tried, at no SI-SDR cost; a sampled, deletion-weighted MWER risk (REINFORCE [10]) adds +0.26 pp in the fusion.
2) *A long-recording stream (12–18 s, the test clips' length) for these losses*, the largest training gain: +1.47 pp on the screening set (two seeds), +0.38 pp on the long set.
3) *A learned per-cell fusion of three model families with a bounded input add-back*, trained through the ASR losses and the post-processing: +0.74 dB SI-SDR over a fixed mix of the same members at equal ΔWER.

Evidence: Tables II and IV.

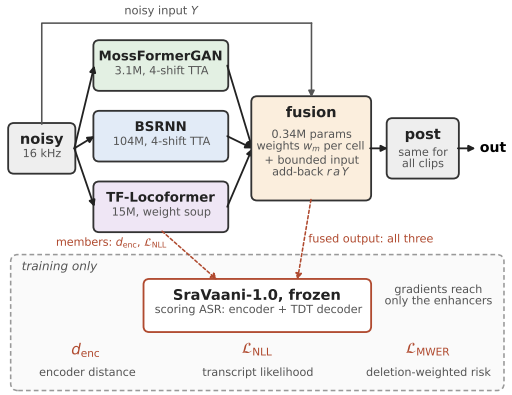


Fig. 1. Every clip takes the same path (top); in training only, the frozen scoring ASR scores members and fusion (bottom).

II. Proposed Method

Preliminary observations. (1) *A cleaner waveform does not give back the words*: on the screening set, at equal SI-SDR (15.8 dB), speech with the events merely turned down 20 dB recovers 92% of the *ceiling* (the ΔWER of the clean reference), our first fixed blend of enhancers only 32%. (2) *The test is longer, not harder*: similar estimated SNR but 13 s clips against 4.5 s, and SraVaani's encoder output for a 2 s crop differs from the same speech inside the full clip (cosine distance 0.24). (3) *Diversity comes from model families, not seeds*: two seeds of a model make nearly the same errors (correlation 0.95–0.99), a new family's errors correlate only 0.69 with our existing ensemble. (4) *Lost words are mostly deletions* (71% then; 89% in the final system).

System overview (Fig. 1). Three enhancers (*members*, Table III) each clean the clip, a small *fusion* network decides in every time-frequency cell how much to trust each, and a fixed post-processing step finishes.

Members and their losses. Each member is fine-tuned with

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 30\, d_{\text{enc}}(\hat{s}, s) + 2.5\, \mathcal{L}_{\text{NLL}} + \lambda\, \mathcal{L}_{\text{MWER}}, \quad (1)$$

plus 15× its compressed-spectrum L1 loss for MossFormerGAN; only the enhancer learns. d_{enc} is the mean per-frame cosine distance between SraVaani's encoder outputs for enhanced and clean speech. SraVaani decodes with a token-and-duration transducer (TDT [11]); $\mathcal{L}_{\text{NLL}}$ is its per-token negative log-likelihood of the target transcript given the enhanced audio, with paths free to start and end within 240 ms of a crop edge. $\mathcal{L}_{\text{MWER}}$ samples four transcripts per window of 8 s or more and raises the likelihood of those with fewer word errors than average (a deletion counts 2; REINFORCE); λ=3 in the fusion and one TF-Locoformer fine-tune, else 0. *Training* (AdamW, cosine decay, a moving average of the weights; exact stages in the repository): MossFormerGAN (2 s crops) and BSRNN (4 s) first learn SI-SDR (and L1), then add the encoder loss, the NLL and the long stream in turn; TF-Locoformer (4 s) trains 20k steps from MERL's checkpoint with all three, then three 4k-step fine-tunes (with MWER, without, louder events) are averaged into one model (a weight soup [12]).

Fusion. Member outputs X_m are scaled to the input Y by least squares (512-point STFT, hop 128). Eight residual blocks of dilated 3×3 convolutions (64 channels) see the log powers of the members, Y and a fixed mix $B = \sum_m w_m^0 X_m$, and each member's phase and level relative to B; per cell they predict softmax weights w_m (from $w^0 = (0.2, 0.35, 0.45)$) and a small complex add-back r of the noisy cell ($|\Re r|, |\Im r| \le 0.2$), allowed only where the mix changed the input:

$$\hat{S} = \sum_m w_m X_m + r\, a\, Y, \quad a = \min\big(1, |Y - B|/|Y|\big). \quad (2)$$

It learns on held-out-protocol copies of the members (486 pairs only) run over the fusion set, the 209 held-out pairs and the screening set, so its inputs are unseen; 3,000 steps with the member losses plus MWER, through the post-processing (its masks held fixed). At inference the all-pair members replace these copies (a swap not measured separately).

Inference and post-processing. MossFormerGAN and BSRNN average over four input delays (0, 41, 83, 124 samples), a test-time augmentation (TTA) known as the shift trick [17]; long clips are decoded in overlapping windows (TF-Locoformer 8 s, BSRNN 20 s, MossFormerGAN 10 s). A light post-processing step changes the fusion output by only −41 dB (median on the test clips): since enhancers alter clean speech slightly even where no event is present, it keeps the model output where it changed the audio (smoothed difference above 1.5% of its maximum, widened 150 ms) and the input elsewhere, mutes stretches 50 dB below the clip level and limits peaks to 0.99 (+0.44 dB, +0.08 pp on the screening set). Every clip gets the same processing; nothing inspects a clip to choose its path.

III. Results and Conclusion

Main results (Table IV). Combined = SI-SDR [18] on synthetic clips + 100 · ΔWER (noisy minus enhanced word error rate, pooled over all 3,178 test clips); local comparisons are paired, on sets the compared models never trained on, with 95% clip-bootstrap intervals. Each enhancer alone already improves both metrics, TF-Locoformer most. The learned fusion adds 0.74 dB SI-SDR over the fixed mix of the same

TABLE II

WHAT WE TRIED, EACH CHANGE AGAINST ITS OWN CONTROL ON A SET THE COMPARED MODELS NEVER TRAINED ON (SCREEN = SCREENING SET; LONG = LONG SET; HELD-OUT = 209 HELD-OUT PAIRS, FUSIONS 2-FOLD OUT-OF-FOLD). ΔWER: CHANGE IN PP [95% CI], ° CI INCLUDES ZERO; ONE SEED UNLESS NOTED. BOARD: CHANGE IN COMBINED OF ONE UPLOAD (NOISE ±0.1). MFGAN = MOSSFORMERGAN, TF-LOCO = TF-LOCOFORMER. MP-SENET: AN EARLY MEMBER, LATER DROPPED. A CHANGE WAS KEPT WHEN ITS ΔWER OR SI-SDR INTERVAL EXCLUDED ZERO AND THE BOARD DID NOT DROP; †KEPT FOR SIMPLICITY AT EQUAL ΔWER, OR ONLY AS ONE OF THREE AVERAGED MODELS.

Change	Model	Set	ΔWER (pp)	SI-SDR (dB)	Board	Kept
Training through the ASR (observations 1, 2, 4)						
Encoder distance, weight 30	MFGAN	screen	+0.30 [0.06, 0.53]	0.00		yes
Encoder distance, weight 30, 2k; 8k steps	MP-SENet	screen	+0.33; +0.54 [0.24, 0.81]	0.00	+0.18 (2k)	yes
+ transcript NLL, weight 2.5	BSRNN 38M; 104M; MFGAN	screen	+0.61 [0.36, 0.85]; +0.43 [0.22, 0.65]; +0.24 [0.03, 0.50]	+0.05; 0.00; 0.00		yes
Long single-recording stream (2 seeds)	BSRNN 38M	screen; long	+1.47 [1.18, 1.78]; +0.38 [0.14, 0.62]	+0.30; –		yes
Long stream also on MFGAN and TF-Loco	3 members	long	+0.33 [0.07, 0.60]	+0.23	+0.11	yes
MWER (weight 3) on a member	TF-Loco	long	−0.17°	–	+0.02	soup†
MWER in the fusion	fusion	long	+0.26 [0.10, 0.43]	−0.02	+0.04	yes
Members, fusion and data (observation 3)						
Learned per-cell fusion vs fixed mix (first version)	fusion	held-out	+0.30°	+0.06 [0.02, 0.10]	+0.13	yes
Add TF-Locoformer (error corr. 0.69 vs 0.82–0.90)	mix	long	–	+0.79 (351/360 clips)	+1.37	yes
Three members instead of eight earlier candidates	fusion	long	−0.04°	−0.11	+0.02	yes†
Weight soup of three fine-tunes	TF-Loco	held-out	+0.17°	+0.085 [0.056, 0.118]	+0.03	yes
198 h more clean speech (IndicVoices-R)	4 families	screen; held-out	+0.10°(MP-SENet)	−0.05 to −0.17		no
Other targets and post-processing						
Keep the input where the model left the audio unchanged	post	screen	+0.08	+0.44	+0.05	yes
Louder output (peak normalisation)	post	screen	+0.71 [0.47, 0.97]	0.00	−0.19	no

TABLE III

PRETRAINED MODELS IN THE FINAL SYSTEM (EXACT VERSIONS IN THE REPOSITORY).

Model (source)	Params	Role
MossFormerGAN (16 kHz speech enhancement), ClearVoice [13]	3.1M	member 1
BSRNN large, trained on VCTK, DNS and WHAMR!, ESPnet [14]	104M	member 2
TF-Locoformer M, trained on DNS 2020, MERL [15]	15M	member 3
SraVaani-1.0, the organisers' ASR [16]	–	frozen loss; transcripts
Silero-VAD [3] and TorchAudio-SQUIM [4]	–	screening clean beds

TABLE IV

MAIN RESULTS, OUT OF SAMPLE (MEMBERS ALONE WITHOUT POST-PROCESSING; FIXED MIX AND FUSION WITH IT), AND THE FINAL BOARD ENTRY.

System	long set SI-SDR (dB)	long set ΔWER (pp)	natural clips ΔWER (pp)
Noisy input	−5.60	0	0
MossFormerGAN alone	13.51	+2.46	−0.06
BSRNN-104M alone	14.49	+2.87	−0.08
TF-Locoformer alone	15.22	+4.13	−0.11
Fixed mix (no learning)	15.30	+3.98	+0.05
Learned fusion (ours)	**16.05**	+4.06 [3.46, 4.76]	−0.05 [−0.38, 0.27]
Speech intact, events at −20 dB	14.39	+6.54	–
Clean speech (the ceiling)	–	+7.59	–
Board, final entry	**15.79**	+11.66 (synthetic)	+0.54
Combined = 15.793 + 4.104 (pooled ΔWER) = **19.897**			

members (both with post-processing) at the same ΔWER, and recovers 53% of the ceiling; the share is the same (52–56%) for clips with little, medium and much of their length under events. On the board, the final entry gains +11.66 pp on synthetic clips, more than locally because the test's synthetic clips lose more words to noise, and +0.54 pp on natural clips, where locally no system changes word error measurably; Combined is 19.897.

Where words are still lost. Of the words our system still loses on test-length clips, 89% are deletions. On the long set, speech left intact under turned-down events keeps 86% of the ceiling at a *lower* SI-SDR than ours, so the limit is setting the right gain in each cell under loud events, not leftover noise. In natural clips, word error is 25.4% in clear speech, 26.8% under local events (2,033 words) and 32.7% under background noise (749 words): removing all noise could gain only about 1 pp.

Validity. The final entry is the best of 45 uploads (history in the repository; only uploads before the organisers' one-pipeline rule handled natural clips separately), so 19.90 is optimistic; one upload resolves only about ±0.1 Combined, so small steps rest on the local intervals of Table II.

Code and checkpoints (with a reproduction guide and every intermediate checkpoint):

https://github.com/mcjoshi/asr-aware-noise-event-removal

https://huggingface.co/manishjoshiai/asr-aware-noise-event-removal

Conclusion. Three enhancers trained against the scoring ASR's own objectives and fused per cell reach Combined 19.90 with every clip processed identically.

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